QUASI-COMPACTNESS DESCENDS ALONG A SURJECTIVE FLAT COVER OF FINITE PRESENTATION OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The definition of a quasi-compact morphism of algebraic stacks through the quasi-compactness of the base changes along morphisms from quasi-compact algebraic stacks. The author's section on morphisms of finite presentation, in which the counted statement occurs. Formulates the assertion to be proved after the base change is known to be quasi-compact.
- The openness on topological spaces of a surjective flat morphism of finite presentation. Invoked inside the counted proof at source line 5969-5970 as a pointer to a lemma of this chapter. Produces a quasi-compact open substack of the base changed cover mapping onto the test stack.

Source reference. Original lemma lemma-fppf-open: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5970–5970. Navigation pointer only: the counted text refers to the Stacks item lemma-fppf-open, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $\mathcal{Z} \to \mathcal{Y}$ be a surjective, flat, locally finitely presented morphism of algebraic stacks. If the base change $\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{Z}$ is quasi-compact, then f is quasi-compact.*

Proof. We have to show that given $\mathcal{Y}' \to \mathcal{Y}$ with $\mathcal{Y}'$ quasi-compact, we have $\mathcal{Y}' \times_{\mathcal{Y}} \mathcal{X}$ is quasi-compact. Denote $\mathcal{Z}' = \mathcal{Z} \times_{\mathcal{Y}} \mathcal{Y}'$. Then $|\mathcal{Z}'| \to |\mathcal{Y}'|$ is open, see Lemma lemma-fppf-open. Hence we can find a quasi-compact open substack $\mathcal{W} \subset \mathcal{Z}'$ mapping onto $\mathcal{Y}'$. Because $\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{Z}$ is quasi-compact, we know that

$$\mathcal{W} \times_{\mathcal{Z}} \mathcal{Z} \times_{\mathcal{Y}} \mathcal{X} = \mathcal{W} \times_{\mathcal{Y}} \mathcal{X}$$

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is quasi-compact. And the map $\mathcal{W} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{Y}' \times_{\mathcal{Y}} \mathcal{X}$ is surjective, hence we win. Some details omitted. $\square$